

Experiment No 8: **Diffie–Hellman Key Exchange (DHKE) algorithm**

Aim:

To simulate the **Diffie–Hellman Key Exchange (DHKE) algorithm** using Python and demonstrate the secure establishment of a **shared secret key** between two communicating users over an insecure communication channel.

Algorithm – Diffie–Hellman Key Exchange

1. Start.
2. Define the prime number p and primitive root g .
3. Read the private key a of User A.
4. Read the private key b of User B.
5. Compute User A's public key:

$$A = g^a \bmod p$$

6. Compute User B's public key:

$$B = g^b \bmod p$$

7. Exchange the public keys A and B .
8. Compute the shared key for User A:

$$K_A = B^a \bmod p$$

9. Compute the shared key for User B:

$$K_B = A^b \bmod p$$

10. Compare both shared keys.
11. If both keys are equal, display "**Secure Session Key Established**".
12. Otherwise, display "**Key Exchange Failed**".
13. Stop.